Finite Certificates and Explicit Sieve Boundaries for Squarefree Quartics and Power-Free Polynomial Values

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Abstract

We examine the local-to-global interfaces in the September 24, 2026 manuscript on squarefree quartics and power-free polynomial values. The finite congruence sieve, rational parameter certificates, and a cyclotomic negative-Pell tail are separated from the source's uniform determinant and geometric counting claims. Exact polynomial enclosure methods from the local Ram corpus provide a reproducible certificate language, not a replacement for arithmetic uniformity. An exact certificate enlarges the first quartic cell from $\eta \geq 1 - 10^{-6}$ to $\eta \geq 249/250$, with a fixed positive post-shift margin; its prime-band consequence keeps all upstream arithmetic and geometric hypotheses. Independently, dyadic negative-Pell uniqueness removes a logarithmic loss from the upper prime tail for $n^4 + 1$. Neither the central tail nor the general squarefree-value density is thereby certified. Contribution assessment: C3/5, explicit enlarged parameter certificates and a log-free cyclotomic tail are proved locally, while the global uniform counting and geometry remain imported obligations.

1 Scope and Proof Architecture

The source manuscript [1] claims the expected density for every fixed irreducible integer polynomial of degree $4 \leq d \leq 8$ with $k = d - 2$, provided no prime kth power divides every value. Its quartic case is squarefreeness. In this report these are *source claims*, not globally certified theorems. All 15 source TeX files have been read in bounded literal displays; the bibliography and README were also read. Reading all these texts is not an independent proof audit of every algebraic-geometric assertion or referenced theorem. The final ledger records exact intervals, hashes, and the private unread complements.

For a fixed $f \in \mathbb{Z}[T]$ and integer $k \geq 2$, write

$$\rho_f(q) = \#\{a \in \mathbb{Z}/q\mathbb{Z} : f(a) = 0 \pmod q\}, \qquad c_f = \prod_p (1 - \rho_f(p^k)/p^k).$$

Negative values are allowed; zero is never power-free. Constants may depend on the fixed polynomial, its degree, and fixed positive margins. No coefficient-height uniformity is asserted.

1.1 What Was Read and What Is Still Imported

The manuscript's dependencies are as follows. Section and line references in this report refer to its local TeX, not PDF pagination.

Source part	Role and retained boundary
Introduction; sieve	Statements, content normalization, finite CRT, medium primes, and transfer from a large-prime tail. These elementary steps are reconstructed below.
Affine counting	Small-height equations from evaluation kernels, smooth patches, residue determinants, mixed-radix records, and random prime deletion. Uniform counting for every moving bounded-degree subvariety remains an imported source assertion.
Arithmetic boxes	Ideal factorization, class correction, unit balancing, and paired charts. Fixed polynomial and finite bad-prime dependence must be retained.
Determinant cuts	Bicone Hilbert polynomial, raw integral determinants, mixed cuts, pure-direction second cuts, and boundary saturation. No claim of independently certifying all these geometric steps.
Lattice boxes	Shortest-vector grouping and polynomial-height unimodular changes; box count X^{m-dj} and weights $u+j+\tau, v+j+\tau$. These are explicit inputs to our conditional summation.
Hilbert; quartic geometry; curves	Thresholds in every dimension, quartic infinite-fiber deletion, degree-two projection alternatives, and one global exceptional quintic-value set. The source's $O_f(X^{1/5})$ exceptional-set count is not reproduced here.
Parameters; certificate	All finite cells and derivative endpoints are independently recomputed. A larger first quartic cell and rational Bernstein witnesses are proved here.
Tail proof	Sum over lattice groups and dyadic prime intervals, with all constants chosen before X. Its upstream inputs are kept named.
Cyclotomic appendix	Exact local factors, Euclidean factorization, Pell upper range, level-equation columns, analytic approximate rank, and box incidence. We sharpen only the Pell range and the first parameter cell; analytic germ nonvanishing remains required.

Several valid details should not be lost in a reformulation. Raw integral determinants, not coefficient-bounded solutions of a moving relation, supply the arithmetic lower bound. The adaptive deletion argument uses a deleted prime independent of the already fixed patch; one cannot condition on stability and then assume independence again. The quintic exceptional set is deleted once globally, not once per box. The analytic appendix gives approximate rank through singular values, not exact rank of analytic evaluations. Content primes and coincident local congruence conditions are retained exactly. Our scoped work found no counterexample to these elementary interfaces; it does not promote the global conclusion to a verified result.

2 Finite Sieve with an Explicit Remainder

Let $I = (A, A+L] \cap \mathbb{Z}$, where $A \in \mathbb{Z}$, $L \in \mathbb{Z}_{\geq 0}$. For a finite set $\mathcal{P}$ of primes put $P = \prod_{p\in\mathcal{P}} p$, $Q = P^k$, and $R_d = \{0 \leq r < d^k : f(r) \equiv 0 \pmod{d^k}\}$. No separability, irreducibility, or good-prime assumption is needed in the following finite statement.

Proposition 2.1 (Classical inclusion-exclusion, exact boundary)**.** *The number $A_{\mathcal{P}}(I)$ of inputs*

avoiding all p^k, $p \in \mathcal{P}$, satisfies

$$A_{\mathcal{P}}(I) = L \prod_{p \in \mathcal{P}} (1 - \rho_f(p^k)/p^k) + E, \tag{1}$$

$$E = \sum_{d \mid P} \mu(d) \sum_{r \in R_d} \left\{ \left\lfloor \frac{A + L - r}{d^k} \right\rfloor - \left\lfloor \frac{A - r}{d^k} \right\rfloor - \frac{L}{d^k} \right\}, \tag{2}$$

$$|E| \le B_{\mathcal{P}} := \prod_{p \in \mathcal{P}} (1 + \rho_f(p^k)) - 1. \tag{3}$$

The exact value in (2) *can be smaller than this envelope.*

Proof. Expand $\prod_{p \mid P}(1 - 1_{p^k \mid f(n)})$ and sum over $n \in I$. Each residue class contributes the difference of the two floors. The Chinese remainder theorem gives $\rho_f(d^k) = \prod_{p \mid d} \rho_f(p^k)$ for squarefree d, so the main term factors. Each floor discrepancy has absolute value less than one. The term $d = 1$ has zero discrepancy because L is integer. The triangle inequality and expansion of the positive product give (3). Empty root sets, content primes, and fixed divisors are included without exceptions. □

This is a classical identity with a more useful displayed error than the source's $O(Q)$, not a new sieve theorem. One may take the minimum of this bound and the elementary period bound Q: write $L = aQ + r$, $0 \le r < Q$, remove a complete periods, and bound the remaining count and its expected value between 0 and r. For fixed separable f of degree d, outside a finite bad set $\rho_f(p^k) \le d$ by simple-root lifting. Thus $B_{\mathcal{P}} \ll_f (d+1)^{\#\mathcal{P}}$, which can be far smaller than $\prod p^k$. This saves no arithmetic tail estimate by itself.

Proposition 2.2 (Quantified interface, not a large-prime theorem)**.** *Suppose f is fixed and separable of positive degree d, and every local factor is positive. Choose integers $2 \le Y \le Z \le X$, where Y contains all bad primes. Let $S_f(X)$ count power-free values in $(X, 2X]$, and set*

$$T_f(X; Z) = \#\{X < n \le 2X : \exists p > Z, p^k \mid f(n)\}.$$

Then, with $\mathcal{P} = \{p \le Y\}$,

$$|S_f(X) - c_f X| \le B_{\mathcal{P}} + \frac{dX}{(k-1)Y^{k-1}} + d\pi(Z) + T_f(X; Z). \tag{4}$$

In particular, $T_f(X; X) = o_f(X)$ suffices for the density by first letting $X \to \infty$ with Y fixed, then letting $Y \to \infty$.

Proof. For $Y < p \le Z$, each of at most d root classes costs at most $X/p^k + 1$. Also $\sum_{p > Y} p^{-k} \le \sum_{m > Y} m^{-k} \le Y^{1-k}/(k-1)$. The positive product satisfies $0 \le c_Y - c_f \le dY^{1-k}/(k-1)$, by the product union bound. Now $S_f \le A_Y$ and $A_Y - S_f$ is at most the medium-prime sum plus T_f. Applying these separately to the upper and lower errors gives (4), with only one copy of the displayed tail sum. Since $\pi(X) = o(X)$, the ordered limit follows. Summing dyadic intervals and bounding the final short prefix transfers the dyadic asymptotic to $[1, N]$. □

Real cutoffs can always be rounded. Even a power-saving bound for $T_f(X; X)$ leaves the elementary $d\pi(X)$ term. Thus improved CRT discrepancy does not supply a power-saving total asymptotic here.

Content and joint restrictions. For $f = \pm cg$ with g primitive and $e = v_p(c) < k$, $\rho_f(p^k) = p^e \rho_g(p^{k-e})$. If $e \geq k$ that prime is a fixed obstruction. Away from the finitely many content primes the root conditions agree. Once Y includes these primes, $0 \leq A_{f,Y} - S_f \leq A_{g,Y} - S_g$, so transfer does not require a new uniform theorem in progressions. For an additional condition $n \bmod M \in \mathcal{A}$, count allowed residues modulo $\operatorname{lcm}(M, Q)$; overlapping conditions cannot be multiplied as independent densities. For products of distinct irreducible factors, finite resultant primes likewise remain in the joint local factor.

3 Exact Polynomial Coordinates from the Private Corpus

Only the following private original intervals were used: the volume16 clearance chapter, lines 62–198, and TR250, lines 155–180 and 455–477. The former proves power-to-Bernstein conversion, convex-hull bounds and exact subdivision; the latter gives finite polynomial moment coordinates and the unisolvence requirement. These are local, confidential sources [2, 3]; no private title or text was sent to a search service. We do not claim to have read the full volumes or full TR250. These finite algebraic tools do not control prime divisibility distributions.

Lemma 3.1 (Finite coordinates and positivity witnesses)**.** *Let $V_r = \mathbb{Q}[t]_{\leq r}$. In the Bernstein basis $B_{j,r}(t) = \binom{r}{j} t^j (1-t)^{r-j}$, a polynomial $p(t) = \sum_{i=0}^{r} \alpha_i t^i$ has coordinates*

$$\beta_j = \sum_{i=0}^{j} \alpha_i \frac{\binom{j}{i}}{\binom{r}{i}}. \tag{5}$$

Consequently $\min_j \beta_j \leq p(t) \leq \max_j \beta_j$ on $[0,1]$. Rational subdivision gives rational certificates on rational intervals. If p is strictly positive on a compact interval, sufficiently fine uniform subdivision certifies positivity by positive coefficients. For any linear coordinates $m_i = \ell_i(p)$ on V_r, equivalence with these coefficients requires the matrix $E_{ij} = \ell_i(t^j)$ to be invertible.

Proof. The identity $\sum_{j=i}^{r} \binom{j}{i} B_{j,r}(t) = \binom{r}{i} t^i$ follows by factoring t^i and applying the binomial theorem. It gives (5); the nonnegative basis functions sum to one. Substituting $t = a + (b-a)s$ preserves rational coefficients, and de Casteljau subdivision consists of rational convex combinations. For fixed degree, coefficients of the restricted polynomial converge uniformly to its value at the left endpoint as the interval length goes to zero: Taylor's formula makes each nonconstant power coefficient $O(b-a)$ on the original compact interval, and the conversion matrix is fixed. A positive minimum therefore proves eventual certification. Finally $m = E\alpha$ proves the last claim. For weighted moments with positive weight almost everywhere, the monomial Gram matrix is positive definite since a nonzero polynomial cannot vanish almost everywhere. Positivity of finitely many quadrature weights alone is insufficient without injective sampling on V_r. □

These are established representation tools, not claimed new identities. Their concrete use below is to issue exact positive-polynomial witnesses for the determinant inequalities throughout a larger admissible cell. A negative Bernstein coefficient means the enclosure is inconclusive, not that the polynomial is negative. Any bounded subdivision routine must return "inconclusive" on exhaustion, not "false theorem".

4 A Larger Quartic Certificate

We retain the source's normalization. For $d \in \{4, 5, 6, 7, 8\}$, $k = d - 2$, a prime interval $[X^\eta, 2X^\eta]$ has cofactor exponent $b = d - k\eta$. A cell $[\eta_-, \eta_+]$ uses $b_- = d - k\eta_+$, $b_+ = d - k\eta_-$. The projective

grid has exponent w, the number of boxes exponent $m = (d-1)w$, and

$$u = (\eta_+ - m)/d, \qquad v = (b_+ - m)/d, \qquad \tau = 1/50000.$$

For $t > 0$ define

$$c_j = \binom{d}{j}\binom{k}{j-1}k^{j-1}, \qquad H(t) = \sum_{j=1}^{d-1} c_j t^j,$$
$$H_a(t) = \sum_{j=1}^{d-1} \frac{d-j}{d} c_j t^j, \qquad H_b(t) = \sum_{j=1}^{d-1} \frac{j}{d} c_j t^{j-1}.$$

These are precisely the normalized source Hilbert coefficients. The two determinant inequalities are

$$w^{d-1}H(t) > \left(\frac{d+1}{d^2}(\eta_+ + b_+ t)\right)^d, \tag{6}$$
$$w^{d-2}\min(H_a(t), H_b(t)) > \left(\frac{\eta_+ + b_+ t}{d-1}\right)^{d-1}. \tag{7}$$

For positive U, V, D, put $\Psi_h(U, V; D) = UV/((kU+V)^{h-1}D)^{1/h}$. The quartic mode uses exactly

$$\delta_4 = \Psi_4(U, V; U+V), \qquad \delta_3 = \Psi_3(U, V; V),$$
$$\delta_2 = \max\{\Psi_2(U, V; \min(3U, 2V, U+V)), U/\sqrt{3}\}, \qquad \delta_1 = \max\{U/2, \min(U, V/4)\}. \tag{8}$$

Set $E_h = \max_{h \le s \le 4} \delta_s$. These are threshold *definitions*. Their validity as arithmetic Hilbert inputs is separately conditional below.

Lemma 4.1 (Source common-shift bound, independently checked)**.** *For the thresholds in* (8)*, and the source's degree-d thresholds with* $4 \le d \le 8$*, if* $3/20 \le V/U \le 5$*, then for every* $s \ge 0$*,*

$$\delta_h(U+s, V+s) \le \delta_h(U, V) + s, \qquad E_h(U+s, V+s) \le E_h(U, V) + s.$$

Proof. It suffices to consider $F = UV/((kU+V)^{h-1}(aU+bV))^{1/h}$ with $(a, b) = (1, 0), (0, 1), (1, 1)$, $2 \le k \le 6$, $2 \le h \le k+2$. The function $UV/(aU+bV)$ is nonnegative, increasing and concave on the positive quadrant: direct differentiation gives a negative semidefinite Hessian (the cases $a = 0$ or $b = 0$ are linear). Write F as the weighted geometric mean of $UV/(kU+V)$ and $UV/(aU+bV)$, with weights $(h-1)/h, 1/h$. The geometric mean is increasing and concave, so F is concave and homogeneous. Writing $f(t) = F(1, t)$, its derivative along $(1, 1)$ at ratio t is $D(t) = f(t) + (1-t)f'(t)$, and $D'(t) = (1-t)f''(t)$. It therefore attains its maximum on $[3/20, 5]$ at an endpoint. At either endpoint,

$$D(t) = \frac{N(t)}{((k+t)^{h-1}(a+bt))^{1/h}}, \quad N(t) = t\left(1 + \frac{1}{t} - \frac{h-1}{h}\frac{k+1}{k+t} - \frac{1}{h}\frac{a+b}{a+bt}\right).$$

All 150 endpoint cases satisfy $N(t) \ge 0$ and $N(t)^h < (k+t)^{h-1}(a+bt)$ by exact rational comparisons. The executable certificate supplies these finite witnesses; the largest strict rational upper root on its 10^{-5} grid is $96031/100000 < 1$. Integrating along the shift proves the branch inequality, since the ratio moves toward one. The $3U$ and $2V$ denominators scale such branches by factors at most one. The remaining linear branches have shift slopes at most one. Finite minima and maxima preserve the increment bound, proving the result. The number $96031/100000$ is *not* a Lipschitz constant for every final threshold: the δ_1 branch U can have slope one. □

Theorem 4.2 (Exact enlarged cell)**.** *For every $0 \le e \le 1/250$, the quartic cell*

$$\eta_- = 1 - e, \quad \eta_+ = 3001/3000, \quad b_- = 2999/1500, \quad b_+ = 2 + 2e$$

admits $t = 1/2$, $w = 199/800$ and $m = 597/800$. It satisfies both strict determinant inequalities, $0 < m < \min(1, \eta_-, b_-)$, and $3/20 \le v/u \le 5$. Uniformly over this family,

$$m + \sum_{h=1}^{4} E_h(u, v) + 4\tau < 1 - \frac{1787}{100000}, \tag{9}$$

$$m + 2u + 2\tau = 1 - \frac{76001}{600000}. \tag{10}$$

At the largest cell $e = 1/250$, $u = 3049/48000$, $v = 5047/16000$ and $v/u = 15141/3049$.

Proof. At $t = 1/2$ the polynomials give $(H, H_a, H_b) = (10, 5, 10)$. The domain and ratio conditions are rational linear comparisons at the endpoints $e = 0, 1/250$. The worst endpoint is bounded by

$$(\delta_4, \delta_3, \delta_2, \delta_1) <_* \left(\frac{1177}{25000}, \frac{507}{10000}, \frac{6901}{100000}, \frac{3049}{48000} \right),$$

where $<_*$ means strict in the first three entries and equality in the fourth at this endpoint. Here $v > 4u$ and $\min(3u, 2v, u + v) = 3u$. The five nonlinear inequalities can be checked simultaneously on the continuous family as follows. Put $x = 250e \in [0, 1]$, $v(x) = (2 - 3w)/4 + x/500$ and $L(x) = 3001/3000 + 1 + x/250$. With $r_4 = 1177/25000$, $r_3 = 507/10000$, $r_2 = 6901/100000$, the following polynomials have positive Bernstein coefficients:

Name	Polynomial in x	Coefficient lower bound
P_1	$10w^3 - (5L/16)^4$	$3/10^6$
P_2	$5w^2 - L^3/27$	$11/1000$
P_4	$r_4^4(2u + v)^3(u + v) - (uv)^4$	$1/10^{10}$
P_3	$r_3^3(2u + v)^2 v - (uv)^3$	$4/10^9$
$P_{2,H}$	$3ur_2^2(2u + v) - (uv)^2$	$8/10^8$

Each bound is strict. Formula (5), or expansion of the displayed rational polynomials, verifies all 21 coefficients; their complete exact values are in the owned test artifact. For an especially short hand check, every nonconstant power coefficient here is negative, so the Bernstein coefficients decrease with their index and the last coefficient is the value at $x = 1$. Also $r_2^2 > u^2/3$. Thus Lemma 3.1 proves all the cut and threshold inequalities on the full interval, not just at a sample of parameters. The cumulative maxima of these bounds sum to $2r_2 + r_3 + r_4 = 1179/5000$; hence $1 - m - (2r_2 + r_3 + r_4) = 359/20000$. Subtracting 4τ gives $1787/100000$. The strict radical bounds give (9). The second identity follows by substitution. □

Full source certificate. Independently implemented integer-power root bracketing and the normalized derivative formulas reproduce every original cell:

d	Cells	Stopping index	Minimum unshifted gap
4	2124	5124	191/100000
5	1084	4084	293/50000
6	552	3552	321/2500
7	226	3226	23177/100000
8	3	3003	31683/100000

The grid minimizer is only a minimizer among $t = i/50$, $1 \le i < 500$, not a certified continuous optimum. Root upper bounds must be strict even at an exact grid root. The new cell is feasible, not asserted maximal. Its width below one is 4000 times the source's 10^{-6} extension, but this does not mean a 4000-fold improvement of an arithmetic exponent or running time.

Corollary 4.3 (Conditional prime-band extension)**.** *Fix a primitive irreducible quartic with positive leading coefficient. Assume the following source inputs for this polynomial: the balanced factorization and paired-chart estimates; the two determinant cuts; the lattice grouping; the uniform weighted Hilbert and curve estimates, including quartic fiber deletion and the one global quintic exceptional set; and the adaptive affine counting theorem for every bounded-degree moving subvariety that occurs. Require all their implied constants to be uniform over the fixed finite charts, degrees and parameter choices, with the choices made before $X \to \infty$. Then there is a $\delta_f > 0$ such that*

$$\#\{X < n \le 2X : \exists \text{ prime } X^{249/250} \le p \le X,\ p^2 \mid f(n)\} \ll_f X^{1-\delta_f}.$$

No global density theorem is asserted unconditionally by this corollary.

Proof. Partition this prime range dyadically. Every such prime tends to infinity, excluding fixed bad primes, and $p^2 \ge X^{249/125} > X$. Thus a root class modulo p^2 contributes at most one integer in an interval of length X. These are precisely the uses of $p > X$ singled out in the source's extension argument (cyclotomic lines 107–120); the representation multiplicity is preserved. Theorem 4.2 supplies uniform coordinate bounds and cut margins on all these intervals. By the named inputs a lattice group has $O_f(X^{m-4j})$ boxes and $U = u + j + \tau$, $V = v + j + \tau$, with $j \ge 0$. Lemma 4.1 gives

$$m - 4j + \sum_h E_h(U, V) \le m + \sum_h E_h(u, v) + 4\tau < 1 - 1787/100000.$$

The fiber cost exponent is $m - 4j + 2(u + j + \tau) \le 1 - 76001/600000$. Fix the affine counting tolerance smaller than half the first margin. Boundedly many groups, charts and components can be summed. Remove the $O_f(X^{1/5})$ exceptional set only once, then absorb the logarithmic number of dyadic intervals by a smaller fixed saving. All constants remain polynomial-specific. This proves the conditional assertion. □

5 Cyclotomic Tail Without a Logarithmic Loss

Unlike Corollary 4.3, the next statement has no determinant or Hilbert input. It sharpens the source's cyclotomic lines 170–191.

Theorem 5.1 (Dyadic negative-Pell uniqueness)**.** *For every integer $D \ge 1$ and real $N \ge 1$, there is at most one positive integer pair (n, q) with $N \le n \le 2N$ and $n^4 + 1 = Dq^2$.*

Proof. If $D = s^2$, then $(sq - n^2)(sq + n^2) = 1$, impossible for $n > 0$. For nonsquare D, put $x = n^2$ and $\alpha = x + q\sqrt{D} = x + \sqrt{x^2 + 1} > 1$. Its norm is -1. Two distinct solutions ordered by $\alpha_2 > \alpha_1$ give

$$\gamma = \alpha_2/\alpha_1 = -\alpha_2\overline{\alpha_1} = a + b\sqrt{D} \in \mathbb{Z}[\sqrt{D}], \qquad a^2 - Db^2 = 1.$$

Here $a = (\gamma + \gamma^{-1})/2$ is an integer greater than one, and $b > 0$ is an integer. The case $a = 2$ would force $Db^2 = 3$, hence $D = 3$, $b = 1$. But $n^4 + 1 \equiv 1$ or $2 \pmod 3$ cannot equal $3q^2$. Thus $a \ge 3$ and $\gamma \ge 3 + \sqrt{8} > 4$. On the other hand the function $\alpha(x) = x + \sqrt{x^2 + 1}$ is strictly increasing and satisfies $\alpha(4x) < 4\alpha(x)$ for $x > 0$, by squaring the positive terms in $\sqrt{16x^2 + 1} < 4\sqrt{x^2 + 1}$. As $N^2 \le x_i \le 4N^2$, the ratio of the two solution values is strictly less than four. This contradiction proves uniqueness. For a fixed n, its positive q is unique, so pairs and inputs agree. □

Corollary 5.2 (Log-free upper tail)**.** *For all real $N \geq 1$, $P > 0$,*

$$\#\{n \in \mathbb{Z} : N \leq n \leq 2N,\ \exists p > P, p^2 \mid n^4 + 1\} \leq \left\lfloor \frac{16N^4 + 1}{P^2} \right\rfloor. \tag{11}$$

The same bound holds if p is allowed to be any positive integer. In particular the bound is at most $17N^{1-2\varepsilon}$ for $P = N^{3/2+\varepsilon}$, $\varepsilon > 0$, and at most $17N/L(N)^2$ for $P = N^{3/2}L(N)$ with $L(N) > 0$.

Proof. Select one divisor p for each input and put $D = (n^4 + 1)/p^2$, a positive integer strictly below $(16N^4 + 1)/P^2$. Distinct inputs cannot yield the same D by Theorem 5.1. Counting the available positive integers D gives (11); including a possible integer endpoint only weakens the bound. Since $N \geq 1$, $16N^4 + 1 \leq 17N^4$. □

This removes the source's $1 + \log N$ loss and permits an $o(N)$ tail above $N^{3/2}L(N)$ for any $L(N) \to \infty$, however slowly. It leaves the central prime range untouched. It uses the exact $n^4 + 1$ equation, not an arbitrary quartic, and does not prove its squarefree-value density. The argument is an elementary retained-range refinement of the source's Pell spacing, not a claimed new Pell theorem in the literature. The valid local root counts are still $\rho_{T^4+1}(4) = 0$ and, for odd p, $\rho_{T^4+1}(p^2) = 4$ exactly when $p \equiv 1 \pmod 8$, and zero otherwise. Thus the candidate Euler product and its elementary lower bound $c_8 > 3/4$ remain valid; equating that product with a density still requires the central tail.

6 Comparison, Limits and Next Research

Current public literature. Public-title searches and primary pages were checked on October 8, 2026. Reuss [4] treats exponent $d - 1$ for general irreducible polynomials; this is not the quartic squarefree exponent 2. Heath-Brown [5] treats the special binomial shape in the range $k \geq (5d+3)/9$, again not $d = 4, k = 2$. Browning's higher-degree result [6] is an imported input in the source, in Xiao's formulation [7]; we have not audited those full proofs. Sofos's 2026 theorem [8], whose introduction and theorem statement were inspected, concerns moments averaged over coefficient boxes with their size related to the input length. It cannot be substituted for a statement about every fixed quartic. The 2026 preprint of Zapata Ceballos–Jalalvand [9] states a more restricted monic degree-$2q$ result under a Galois subfield and local condition. Its statement and finite-sieve discussion were inspected; its global claim is not an input here. A positive product of finite congruence densities alone is not an estimate for the unsieved tail. The searches are a scoped comparison, not an exhaustive novelty search or a declaration of community acceptance of the September source.

Corrections versus refinements. No false numerical entry was found in the source's full certificate. The changes proved here are stronger local estimates, not corrections to false source inequalities: the finite CRT error becomes explicit, the first quartic cell becomes wider, and the cyclotomic Pell bound loses its unnecessary logarithm on a dyadic interval. The source's distinction between a power-saving large-prime proposition and an $o(X)$ final density is valid and retained. We do not identify an established fatal gap in its global theorem from these finite tests, and we do not claim to have ruled out such a gap. In particular, arithmetic uniformity in the adaptive counting theorem and geometry for every moving subvariety cannot be certified by a parameter table.

Advantages and costs. Rational witnesses are portable, strict endpoint roots are handled correctly, and the same polynomial coefficients can be represented by unisolvent finite moments

without losing information. The Bernstein form gives a continuous certificate rather than an empirical grid check. The larger cell leaves a visibly positive tolerance for summing lattice groups. However, it does not lower determinant degrees automatically, does not optimize the full parameter space, and does not supply coefficient-height uniformity. The exact inclusion-exclusion formula has exponentially many terms in the number of small primes; its practical use is a bound or a small finite calculation, not a large-scale sieve algorithm. The Pell refinement is unconditional and elementary, but specific to the constant $+1$ equation.

Feasible next steps. The next bounded numerical task is rational interval optimization of (e, t, w), retaining all threshold branches and strict margins; failed positivity enclosures must remain inconclusive. A more substantive mathematical task is to isolate the adaptive affine theorem with explicit degree/error dependencies and verify the hypotheses for all its moving strata, not just initial cut components. Any improvement to the full density error also needs a better medium-prime treatment than $d\pi(X)$. The Pell argument suggests studying which other fixed norm equations have a cofactor-uniform multiplicative spacing, but neither arbitrary quartics nor arbitrary right-hand constants inherit Theorem 5.1 without a new proof.

7 Reproducibility and Assessment

The owned standard-library program `_audit/M046-local-checks.py` is run with `-full`. It does not execute the source's embedded program. It recomputes its normalized polynomials, enumerates all 499 bidegree-ratio choices per cell, uses integer comparisons for strict rational upper roots, and matches the five stopping indices and exact minimum gaps. Separate checks cover 150 derivative endpoints, 700 exact-root grid cases, the five continuous positivity polynomials, and finite CRT counts including content and fixed-square obstructions. The Pell tests enumerate $1 \le D \le 200$, $1 \le n \le 500$, then check 250 dyadic windows for each D; most such windows are empty. Additional negative-Pell solutions with $D \le 100, q \le 1000$ check the integer ratio step. These manufactured finite checks supplement, not replace, the universal proofs above. Parent checks are not included in the author count or frozen as owned evidence.

The exact run counts, all rational witnesses, source-read ledger, local proof map, literature scope, native/export compilation results, page-by-page visual review, and hashes are supplied in the owned audit files. The private intervals total 186 distinct lines, not entire books. All source scientific TeX was literally read, while imported reference proofs, the Lean formalization and private unread complements remain outside the claimed audit. The source snapshot provenance is handled by the parent; the author independently checks local file identities.

The demonstrated contribution is **C3/5**, separately from the unchanged catalogue applicability **L3**. It rests on Theorem 4.2, its conditional consequence Corollary 4.3, and Theorem 5.1 with Corollary 5.2. The classical CRT identity, Bernstein coordinates and source common-shift method are not credited as novel theorems. The concrete source-specific range and tail improvements are more than relabeling a coordinate system, but they do not replace the global arithmetic estimates and therefore do not justify C4 or C5.

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